

# BMI HEALTH ANALYTICS DASHBOARD REPORT

*A Power BI Business Intelligence Case Study*

*Figure: BMI Health Analytics Dashboard – Power BI*

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## 2. Executive Summary

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The BMI Health Analytics Dashboard is a Power BI-based business intelligence solution designed to consolidate, visualize, and interpret health and anthropometric data collected across a sample population. The dashboard integrates Body Mass Index (BMI), body composition, activity levels, and dietary indicators into a single interactive interface, enabling stakeholders such as healthcare administrators, wellness program managers, and academic researchers to monitor population health trends efficiently.

The primary business objective of the dashboard is to support data-driven decision-making in the domain of preventive healthcare and wellness management. By aggregating key metrics such as Maximum BMI, Minimum BMI, Average Weight, Average Height, Average Body Fat Percentage, and Average Daily Calorie Intake, the dashboard provides a rapid, at-a-glance assessment of the overall health profile of the monitored population.

Analysis of the dashboard reveals several major findings. The population BMI values range widely, from a minimum of 10.02 to a maximum of 60.82, indicating the coexistence of severely underweight and severely obese individuals within the dataset. The Medium health risk category and the Normal Weight BMI class collectively represent the largest segments of the population, while a substantial proportion of individuals fall into Obese Class categories, signalling a considerable public health concern. A clear positive relationship is observed between age and average body fat percentage, and daily calorie intake patterns vary meaningfully across BMI classifications.

Overall, the dashboard offers strong analytical value for decision-making. Its interactive filters for Gender, Activity Level, BMI Class, and Health Risk Level allow users to isolate specific population segments and drill into targeted health patterns. This capability supports the design of personalized wellness interventions, targeted nutritional counselling programs, and early-warning health risk monitoring systems. The dashboard, therefore, functions not merely as a reporting tool but as a strategic decision-support instrument for healthcare and wellness stakeholders.

## 3. Introduction

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### 3.1 What is BMI?

Body Mass Index (BMI) is a widely used screening metric that estimates a person's body fat based on their height and weight. It is calculated by dividing an individual's weight in kilograms by the square of their height in metres. BMI values are typically classified into standard categories such as Underweight, Normal Weight, Overweight, and various classes of Obesity, providing a simple numerical indicator of an individual's weight status relative to their height.

### 3.2 Why BMI Analysis is Important

BMI analysis serves as a foundational tool in public health and clinical assessment because it offers a low-cost, non-invasive method of identifying individuals who may be at risk of weight-related health conditions. Sustained deviations from a healthy BMI range are statistically associated with increased risk of cardiovascular disease, type 2 diabetes, musculoskeletal disorders, and other chronic conditions.

Consequently, systematic BMI analysis at the population level enables healthcare organizations to prioritize resources and design targeted interventions.

### 3.3 Importance of Monitoring Health Risks

Continuous monitoring of health risk indicators allows organizations to detect adverse trends before they escalate into serious clinical outcomes. By tracking health risk levels alongside BMI, body fat percentage, and activity data, decision-makers can identify high-risk subgroups within a population and allocate preventive resources accordingly, thereby reducing long-term healthcare costs and improving overall wellbeing.

### 3.4 Role of Data Analytics in Healthcare

Data analytics has become an integral component of modern healthcare management. Business intelligence platforms such as Power BI enable the transformation of raw health records into interactive, visual dashboards that support rapid interpretation and evidence-based decision-making. Analytics tools facilitate the identification of patterns, correlations, and outliers within health datasets that would be difficult to detect through manual review, thereby enhancing the speed and accuracy of healthcare-related decisions.

### 3.5 Purpose of Developing this Dashboard

The BMI Health Analytics Dashboard was developed to provide a consolidated, interactive platform for exploring the relationships between demographic attributes, body composition, activity levels, and dietary intake within a health dataset. The dashboard is intended to support academic analysis, internship-based project work, and business intelligence portfolio development, while also demonstrating practical application of Power BI, Power Query, and DAX in the healthcare analytics domain.

## 4. Dashboard Overview

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The dashboard is organized into a single-page interactive layout that combines summary KPI cards, filtering controls, and multiple chart types. This design allows users to obtain both a high-level summary and a granular, segment-level view of the underlying health dataset without navigating across multiple screens.

### 4.1 Dashboard Components

- **KPI Cards:** Six vertically stacked cards on the left panel display Maximum BMI, Average Height, Minimum BMI, Average Weight, Average Body Fat Percentage, and Average Daily Calorie Intake.
- **Filters (Slicers):** Interactive controls for Gender, Activity Level, BMI Class, and Health Risk Level positioned along the top of the dashboard.
- **Stacked Column Chart:** “BMI Distribution by Health Risk and Gender”, displaying the count of BMI classifications across health risk levels, segmented by gender.
- **Pie Chart:** “BMI Class Distribution”, showing the proportional share of each BMI classification within the population.

- Line Chart: “Average Body Fat (%) by Age”, illustrating the trend of body fat percentage as age increases.
- Treemap: “Average Daily Calorie Intake by BMI Class”, representing relative calorie intake across BMI categories using proportionally sized rectangles.
- Sankey Diagram: “Weight by Gender and Activity Level”, visualizing the flow of weight distribution across gender and activity level segments.

4.2 User Interaction

Users interact with the dashboard primarily through the slicers located at the top of the report. Selecting a specific Gender, Activity Level, BMI Class, or Health Risk Level dynamically filters all visuals simultaneously, allowing users to explore how KPIs and chart patterns change for specific population segments. This cross-filtering behaviour, native to Power BI, enables rapid, exploratory analysis without requiring technical query-writing skills.

5. Dataset Overview

The dataset underlying the dashboard captures anthropometric, demographic, and lifestyle attributes for each individual in the sample population. Based on the fields referenced and visualized within the dashboard, the following attributes are present in the underlying data model.

| Field               | Description / Significance                                                                                           |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| Gender              | Categorical attribute (Female, Male, Other) used to segment BMI, weight, and activity patterns by demographic group. |
| Age                 | Numeric attribute used as the horizontal axis in the body fat trend analysis; enables age-based health monitoring.   |
| Height              | Numeric measurement (cm) used in conjunction with Weight to derive BMI; also reported as an average KPI.             |
| Weight              | Numeric measurement (kg) used to compute BMI and analysed by Gender and Activity Level in the Sankey diagram.        |
| BMI                 | Calculated index used to summarize Maximum and Minimum values at the population level.                               |
| BMI Class           | Categorical classification (Underweight, Normal Weight, Overweight, Obese Class 1/2/3, Other) derived from BMI.      |
| Body Fat Percentage | Numeric attribute reflecting body composition; averaged as a KPI and trended against Age.                            |
| Health Risk Level   | Categorical attribute (Low, Medium, High, Severe, Very High) indicating relative health risk severity.               |
| Activity Level      | Categorical attribute (e.g., Active, Lightly Active, Moderately Active) used to segment the Sankey diagram.          |

| Field                | Description / Significance                                                                              |
|----------------------|---------------------------------------------------------------------------------------------------------|
| Daily Calorie Intake | Numeric attribute representing average caloric consumption, analysed across BMI classes in the treemap. |

Together, these attributes allow the dashboard to connect demographic characteristics with physiological measurements and behavioural indicators, forming a comprehensive basis for health analytics.

## 6. Key Performance Indicators (KPIs)

Six KPI cards summarize the overall health profile of the dataset at a glance. The table below presents each KPI along with its observed value and business interpretation.

| KPI                          | Value          | Business Interpretation                                                                                                                                                    |
|------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum BMI                  | 60.82          | Indicates the presence of at least one individual in the extreme obesity range, signalling a critical health risk case requiring clinical attention.                       |
| Average Height               | 169.92 cm      | Represents the mean stature of the population, providing context for interpreting weight and BMI figures across the dataset.                                               |
| Minimum BMI                  | 10.02          | Reflects the presence of a severely underweight individual, indicating potential malnutrition or an underlying health condition requiring intervention.                    |
| Average Weight               | 80.16 kg       | Serves as a baseline body-mass reference; when viewed alongside average height, suggests a population BMI skewed toward the higher end of the healthy-to-overweight range. |
| Average Body Fat             | 38.81%         | A relatively elevated average body fat percentage across the population, suggesting that a meaningful share of individuals carry higher-than-recommended adiposity levels. |
| Average Daily Calorie Intake | 2.61K Calories | Provides a benchmark for nutritional intake; when cross-referenced with BMI Class, helps identify whether calorie consumption aligns with recommended dietary guidelines.  |

Collectively, these KPIs indicate a population with considerable variability in health status — ranging from severe underweight to extreme obesity — and an average body composition profile that warrants targeted wellness intervention.

## 7. Dashboard Visual Analysis

### A. BMI Distribution by Health Risk and Gender

Chart Type: Stacked Column Chart. Purpose: To display the count of individuals across each Health Risk Level (Medium, Low, High, Severe, Very High), broken down by Gender (Female, Male, Other).

### Observations

- The Medium health risk category contains the largest population count, substantially exceeding all other categories.
- The Low, High, and Severe categories display comparatively similar, moderate population counts.
- The Very High health risk category has the smallest population share, though it remains a non-trivial group requiring attention.
- Female and Male segments appear roughly comparable in most columns, with the Other gender category consistently representing a smaller proportion.

### Business Insights

- The concentration of individuals in the Medium risk category suggests that the population is not predominantly at extreme risk, but a large “watch list” segment exists that could migrate toward higher risk without intervention.
- The presence of a Very High risk segment, however small, represents a priority group for clinical follow-up and case management.

### Key Findings

- Health risk is not evenly distributed; it is heavily skewed toward the Medium category.

### Possible Actions

- Design early-intervention wellness programs targeted specifically at the Medium risk segment to prevent progression to higher risk categories.
- Establish a clinical monitoring protocol for individuals in the Severe and Very High risk categories.

## B. BMI Class Distribution

Chart Type: Pie Chart. Purpose: To present the proportional share of each BMI classification (Normal Weight, Overweight, Underweight, Obese Class 1, Obese Class 2, Obese Class 3, Other) within the total population.

### Observations

- Normal Weight represents the largest single segment at approximately 28.64% (3.47K individuals) of the population.
- Overweight accounts for approximately 16.66% (2.02K individuals).
- Obese Class 1 accounts for approximately 16.47% (1.99K individuals), while Obese Class 3 and Obese Class 2 account for approximately 14.24% (1.72K) and 9.59% (1.16K) respectively.
- Underweight and Other categories together account for the remaining share, with Underweight representing roughly 14% of the population.

### Business Insights

- When the three Obese classes are combined, obese individuals collectively constitute a substantial proportion of the population — comparable in scale to the Normal Weight segment — highlighting obesity as a significant population health concern.
- The Underweight population, while smaller, remains large enough to warrant dedicated nutritional support programs.

### Key Findings

- The population is not weight-homogeneous; it spans the full spectrum from underweight to severe obesity, with obesity-related classes collectively rivaling the Normal Weight category.

### Possible Actions

- Segment obesity-focused interventions by class severity, prioritizing Obese Class 3 individuals for immediate clinical engagement.
- Introduce targeted nutritional counselling for the Underweight segment to address potential malnutrition risk.

## C. Average Body Fat (%) by Age

Chart Type: Line Chart. Purpose: To illustrate the trend in average body fat percentage as age increases from approximately 0 to 50+ years.

### Observations

- The line trend shows a consistent upward trajectory, rising from approximately 35% body fat at younger ages to approximately 45% at older ages.
- Minor fluctuations (noise) are present throughout the trend, but the overall directional pattern remains clearly positive.

### Business Insights

- The steady increase in body fat percentage with age aligns with well-documented physiological patterns, where metabolic rate typically declines and fat accumulation increases with advancing age.

### Health Implications

- Older age groups within the dataset may face elevated risk of obesity-related conditions such as cardiovascular disease and type 2 diabetes, warranting age-specific preventive health strategies.

### Possible Actions

- Introduce age-tiered wellness checkpoints, with increased frequency of body composition assessment for individuals above the midpoint of the observed age range.

## D. Average Daily Calorie Intake by BMI Class

Chart Type: Treemap. Purpose: To compare average daily calorie intake across BMI classifications, with rectangle size proportional to relative calorie intake.

### Observations

- Obese Class 3 and Obese Class 1 occupy the largest rectangles, indicating these groups report the highest average daily calorie intake among all BMI classes.
- Obese Class 2 and Overweight occupy moderately sized rectangles, reflecting comparatively high but slightly lower intake levels.
- The Other and Normal Weight categories occupy mid-sized rectangles, while Underweight occupies the smallest rectangle, indicating the lowest average calorie intake among all classes.

### Business Insights

- A clear positive association is observable between higher BMI classification and higher average daily calorie intake, suggesting that caloric consumption patterns are a contributing factor to elevated BMI within this dataset.
- The comparatively low calorie intake within the Underweight class supports the hypothesis that insufficient caloric consumption is linked to low BMI in this population.

### Nutritional Interpretation

- These findings reinforce the importance of calorie-balanced dietary planning as a core component of weight management programs across all BMI classes.

### Possible Actions

- Deploy structured calorie-reduction and portion-control guidance for Obese Class 1–3 segments.
- Develop calorie-enrichment nutritional plans for the Underweight segment.

## E. Weight by Gender and Activity Level

Chart Type: Sankey Diagram. Purpose: To visualize how total weight is distributed and flows across Gender categories (Other, Female, Male) and Activity Level categories (Active, Lightly Active, Moderately Active, and others).

### Observations

- Weight flows are distributed across all three gender categories, with Female and Male segments showing broader flow bands compared to the Other category, indicating larger sample representation.
- Each gender category branches into multiple activity level segments, with Moderately Active and Lightly Active flows appearing prominently across all genders.

### Gender Differences

- The relative width of flow bands suggests differences in the distribution of activity levels between Female and Male segments, which may correspond to differing average weight profiles between genders.

### Activity Level Influence

- The diagram indicates that activity level is not uniformly distributed within gender groups, implying that lifestyle and activity patterns vary meaningfully across demographic segments.

### Key Relationships

- The Sankey structure reveals that weight outcomes are influenced by the interaction between gender and activity level rather than either factor in isolation.

Possible Actions

- Design gender-responsive activity programs that account for observed differences in activity-level distribution.
- Encourage transition of individuals from lower activity categories toward more active lifestyle segments through structured fitness initiatives.

8. Dashboard Filters

The dashboard incorporates four interactive slicers that allow users to dynamically narrow the scope of analysis. Each filter serves a distinct analytical purpose.

| Filter            | Purpose                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gender            | Enables users to isolate and compare health metrics (BMI, weight, body fat) between Female, Male, and Other demographic segments.                             |
| Activity Level    | Allows users to examine how physical activity (e.g., Active, Lightly Active, Moderately Active) correlates with weight and BMI outcomes.                      |
| BMI Class         | Enables focused analysis of a specific weight category, such as isolating only Obese Class 3 or Underweight individuals for detailed review.                  |
| Health Risk Level | Allows users to filter the dashboard to display only individuals within a specific risk tier (e.g., Severe or Very High), supporting prioritized case review. |

The ability to combine these filters simultaneously significantly improves data exploration and decision-making. For example, a healthcare administrator could filter the dashboard to display only Female individuals classified as Obese Class 2 with a Severe health risk level, instantly generating a focused view of a specific high-priority sub-population without requiring any additional query development. This self-service analytical capability reduces dependency on technical teams and accelerates the pace of evidence-based decision-making.

9. Key Insights

Based exclusively on the visual evidence presented in the dashboard, the following analytical insights have been identified:

1. The population BMI range spans from 10.02 to 60.82, reflecting extreme variability from severe underweight to extreme obesity.
2. The Medium health risk category contains the largest share of the population, indicating a large at-risk group that has not yet progressed to severe risk.
3. The Very High health risk category, while the smallest segment, represents a critical group requiring immediate clinical attention.

4. Normal Weight is the single largest BMI class at approximately 28.64% of the population.
5. Combined Obese Class 1, 2, and 3 categories collectively represent a share of the population comparable in scale to the Normal Weight segment.
6. Average body fat percentage across the population stands at 38.81%, a relatively elevated figure suggesting widespread excess adiposity.
7. Average body fat percentage increases consistently with age, rising from approximately 35% to 45% across the observed age range.
8. Average daily calorie intake (2.61K calories) is highest among Obese Class 1 and Obese Class 3 individuals, as shown in the treemap visualization.
9. The Underweight BMI class corresponds to the lowest average daily calorie intake among all classes, consistent with expected nutritional patterns.
10. Average weight (80.16 kg) combined with average height (169.92 cm) suggests the overall population BMI skews toward the overweight range.
11. Gender segmentation reveals that Female and Male groups are the dominant contributors to the dataset, with the Other category present but comparatively smaller.
12. Activity level appears to meaningfully influence weight distribution, as illustrated by the branching structure of the Sankey diagram.
13. The coexistence of both severe underweight and severe obesity cases within the same dataset indicates a bimodal health risk profile requiring differentiated intervention strategies.
14. The stacked column chart indicates that health risk levels do not vary drastically by gender, suggesting risk factors may be more strongly associated with BMI and lifestyle than with gender alone.
15. The proportional sizing of the treemap confirms a direct relationship between BMI severity and caloric intake, reinforcing diet as a key lever for weight management.
16. The dashboard's multi-dimensional filtering (Gender, Activity Level, BMI Class, Health Risk Level) enables granular sub-population analysis not achievable through static reporting.
17. The relatively small size of the Obese Class 2 segment compared to Obese Class 1 and 3 suggests a non-linear distribution across obesity sub-classes that may merit further investigation.

## 10. Business Recommendations

The following actionable recommendations are proposed on the basis of the patterns observed within the dashboard:

18. Launch structured healthy eating initiatives targeting individuals in Obese Class 1 through Class 3, with emphasis on calorie-balanced meal planning.
19. Develop supervised exercise programs tailored to Lightly Active and Moderately Active segments to encourage progression toward higher activity levels.
20. Design personalized wellness plans that combine BMI class, activity level, and age data to create individualized health improvement pathways.

21. Implement continuous BMI monitoring protocols, particularly for individuals near classification boundaries who are at risk of moving into a higher-risk category.
22. Establish early intervention programs for individuals newly identified within the Medium health risk category to prevent escalation to Severe or Very High risk.
23. Provide nutritional counselling services for both the Underweight and Obese population segments, addressing malnutrition and overconsumption respectively.
24. Prioritize obesity prevention campaigns given the substantial combined share of Obese Class 1–3 individuals within the population.
25. Introduce workplace wellness campaigns that incorporate BMI screening, activity tracking, and nutritional education for employee populations.
26. Conduct health awareness programs focused on the relationship between age and body fat accumulation, encouraging proactive lifestyle adjustments in older age groups.
27. Schedule regular fitness assessments, particularly for individuals classified under Severe and Very High health risk levels, to track progress and adjust intervention plans.
28. Allocate additional clinical resources toward the Very High health risk segment given its critical priority status despite its smaller population size.
29. Use the Activity Level and Gender segmentation from the Sankey diagram to design gender-responsive fitness engagement strategies.

## 11. Dashboard Strengths

- **Interactive Filtering:** Four independent slicers (Gender, Activity Level, BMI Class, Health Risk Level) allow rapid, self-service exploration of population sub-segments.
- **Easy Interpretation:** KPI cards present headline figures in a clear, immediately readable format suitable for non-technical stakeholders.
- **KPI Monitoring:** The six summary cards provide a concise, at-a-glance snapshot of the overall health profile of the population.
- **Data Visualization Effectiveness:** The use of varied chart types — stacked column, pie, line, treemap, and Sankey — enables multi-dimensional analysis of the same underlying dataset from different analytical perspectives.
- **User-Friendly Layout:** The single-page dashboard layout consolidates all key metrics and visuals without requiring navigation across multiple report pages.
- **Decision Support Capability:** The combination of KPIs, trend analysis, and categorical breakdowns equips decision-makers with sufficient context to prioritize interventions effectively.

## 12. Limitations

- **Limited Demographic Variables:** The dataset does not appear to include additional demographic factors such as income level, education, or occupation that could enrich health risk analysis.

- **Lack of Historical Trends:** The dashboard reflects a single point-in-time snapshot and does not track how KPIs or population health metrics have changed over time.
- **No Geographic Analysis:** The dashboard does not include location-based segmentation, limiting the ability to identify regional health disparities.
- **Static Dataset:** The underlying data does not appear to be refreshed in real time, which may limit its relevance for ongoing operational decision-making.
- **No Predictive Analytics:** The dashboard is descriptive in nature and does not currently forecast future BMI trends or health risk trajectories.
- **Missing Clinical Parameters:** Additional clinical indicators such as blood pressure, cholesterol levels, or blood glucose are not included, which would provide a more complete health risk picture.

## 13. Future Enhancements

- **Predictive BMI Modelling:** Incorporate statistical or machine learning models to forecast an individual's future BMI trajectory based on current trends.
- **Machine Learning for Health Risk Prediction:** Apply classification algorithms to proactively predict which individuals are likely to migrate into higher health risk categories.
- **Real-Time Health Data Integration:** Connect the dashboard to live data sources to enable continuous, up-to-date monitoring rather than static snapshots.
- **Wearable Device Connectivity:** Integrate data from fitness trackers and wearable devices to enrich activity level and calorie expenditure accuracy.
- **Geographic Mapping:** Introduce location-based visualizations to identify regional variation in BMI and health risk patterns.
- **Time-Series Analysis:** Add historical trend visualizations to track how KPIs and population health metrics evolve over multiple time periods.
- **Personalized Health Recommendations:** Develop a recommendation engine that generates individualized wellness suggestions based on a person's specific health profile.
- **Mobile Dashboard Support:** Optimize the dashboard layout for mobile devices to improve accessibility for field-based healthcare staff.
- **AI-Powered Health Alerts:** Implement automated alerting mechanisms that notify administrators when an individual's metrics cross critical risk thresholds.

## 14. Conclusion

The BMI Health Analytics Dashboard successfully achieves its core objective of consolidating anthropometric, demographic, and lifestyle data into a single, interactive business intelligence tool. Through its combination of KPI cards, categorical breakdowns, trend analysis, and flow diagrams, the dashboard provides a comprehensive view of population health status spanning the full spectrum from severe underweight to extreme obesity.

The key findings of this analysis — including the wide BMI range, the elevated average body fat percentage, the positive relationship between age and body fat, and the association between BMI class and caloric intake — collectively demonstrate the analytical depth achievable through well-designed data visualization. These findings carry direct business value, informing the design of targeted wellness programs, nutritional interventions, and risk-based clinical prioritization strategies.

In the broader context of healthcare applications, this dashboard illustrates how business intelligence tools can bridge the gap between raw health data and actionable organizational decisions. It reinforces the importance of data-driven decision-making as a cornerstone of modern preventive healthcare management, and provides a strong foundation for future enhancement through predictive analytics, real-time integration, and personalized health recommendation systems.

## 15. Appendix

### 15.1 Dashboard Features

- KPI Cards for rapid summary reporting
- Interactive Filters (Slicers) for dynamic, multi-dimensional exploration
- Multiple Chart Types for varied analytical perspectives
- Responsive, single-page design for consolidated viewing

### 15.2 Visualization List

- Stacked Column Chart – BMI Distribution by Health Risk and Gender
- Pie Chart – BMI Class Distribution
- Line Chart – Average Body Fat (%) by Age
- Treemap – Average Daily Calorie Intake by BMI Class
- Sankey Diagram – Weight by Gender and Activity Level

### 15.3 Tools Used

- Power BI – Dashboard design and interactive visualization
- Power Query – Data extraction, transformation, and loading (ETL)
- DAX (Data Analysis Expressions) – Calculated KPIs and measures
- Microsoft Excel – Supplementary data preparation and validation